Digital Logic - 500+ One-Liners

Coal India Limited - Management Trainee (Systems) | CBT Preparation

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Number Systems

- There are **4** number systems in digital electronics: **binary (base 2)**, **octal (base 8)**, **decimal (base 10)**, and **hexadecimal (base 16)**.
- Binary uses **2** symbols: **0** and **1**.
- Octal uses **8** symbols: **0 through 7**.
- Decimal uses **10** symbols: **0 through 9**.
- Hexadecimal uses **16** symbols: **0-9 and A-F**, where A-F represent **10,11,12,13,14,15**.
- The largest digit in base r is **r-1**, so the largest binary digit is **1** and the largest octal digit is **7**.
- A group of **4** bits is called a **nibble**, and **8** bits form a **byte**.
- One hexadecimal digit maps to exactly **4** binary bits, while one octal digit maps to exactly **3** binary bits.
- The **MSB** is the leftmost bit with the highest weight, and the **LSB** is the rightmost bit with weight **2^0 = 1**.
- The value of a positional number equals **Sigma (digit x base^position)**, summed over all positions.
- There are **2** methods to convert decimal to another base: **repeated division** for the integer part and **repeated multiplication** for the fractional part.
- In decimal-to-base integer conversion, remainders are read **bottom-to-top**.
- In decimal-to-base fractional conversion, carries (integer parts) are read **top-to-bottom**.
- To convert binary to octal you group bits in **3s**, and to convert binary to hex you group bits in **4s**, starting from the **radix point**.
- To convert octal to binary expand each digit to **3 bits**, and for hex to binary expand each digit to **4 bits**.
- The safest octal-to-hex path is **via binary**, regrouping 3-bit sets into 4-bit sets.
- The example **(1011.101)_2** equals **(11.625)_10** in decimal.
- The example **(25)_10** equals **(11001)_2** in binary.
- The example **(0.625)_10** equals **(0.101)_2** in binary.
- For binary integer grouping you pad with **leading zeros**, but for fractional grouping you pad with **trailing zeros**.
- Some decimal fractions like **0.1** are **non-terminating** in binary.

Binary Arithmetic

- In binary addition **1+1 = 10**, giving sum **0** and carry **1**.
- In binary addition **1+1+1 = 11**, giving sum **1** and carry **1**.
- In binary subtraction **0-1 = 1** with a **borrow of 1**.
- Multiplying a binary number by **2^n** shifts it **left** by n positions.
- Dividing a binary number by **2^n** shifts it **right** by n positions.
- Binary multiplication uses **shift-and-add**, and binary division uses **shift-and-subtract**.
- The example **1011 + 0110** equals **10001** (11+6=17).
- The example **1101 x 101** equals **1000001** (13x5=65).

Signed Representation

- There are **3** signed-number schemes: **sign-magnitude**, **1's complement**, and **2's complement**.
- In all signed schemes the **MSB** is the sign bit, where **0** means positive and **1** means negative.
- Sign-magnitude has **2** representations of zero: **+0** and **-0**.
- 1's complement also has **2** zeros and uses an **end-around carry** during addition.
- 2's complement has exactly **1** representation of zero and is the **modern standard**.
- The 1's complement is formed by **inverting all bits**, and 2's complement by **inverting all bits and adding 1**.
- The shortcut for 2's complement is to copy bits up to the **first 1** and **invert** the rest.
- The n-bit unsigned range is **0 to 2^n - 1**, giving **2^n** values.

- The n-bit 2's-complement signed range is **-2^(n-1)** to **+2^(n-1) - 1**.
- An 8-bit 2's-complement value ranges from **-128** to **+127**.
- A 4-bit 2's-complement value ranges from **-8** to **+7**.
- A 16-bit unsigned maximum is **65535**, and 16-bit signed max is **+32767**.
- Only **2's complement** has an asymmetric range with one extra **negative** value.
- To subtract A-B in 2's complement, add the **2's complement of B** to A and **discard** the carry-out.
- If 2's-complement subtraction produces **no carry-out**, the result is **negative**.
- In 1's-complement subtraction a carry-out triggers an **end-around carry** (add 1 to LSB).
- **Overflow** occurs only when **two same-sign operands** produce an **opposite-sign** result.
- Adding numbers of **opposite signs** can **never** cause overflow.
- The hardware overflow rule is **overflow = Cin(MSB) XOR Cout(MSB)**.
- The example **0111 + 0001** in 4-bit signed gives **1000**, an **overflow** (+7+1 = -8).
- A **carry-out** is not the same as **overflow**; carry-out is often discarded in 2's complement.

Codes

- **BCD (8421)** encodes each decimal digit in **4** bits using weights **8-4-2-1**.
- In BCD the codes **1010 to 1111** (10-15) are **invalid**.
- **Packed BCD** stores **2** decimal digits per byte, **4 bits** each.
- BCD addition adds a correction of **6 (0110)** when a nibble exceeds **9** or generates a carry.
- Decimal **59** in BCD is **0101 1001**, differing from pure binary **111011**.
- BCD is a **weighted** code but is **less storage-efficient** than binary.
- **Excess-3** is formed by **adding 3** to the BCD value of each digit.
- Excess-3 is **non-weighted** yet **self-complementing** (its 9's complement equals its 1's complement).
- Decimal **5** in Excess-3 is **1000** (5+3=8).
- **Gray code** is a **non-weighted, reflected, cyclic** code where only **1 bit** changes between consecutive values.
- Binary-to-Gray copies the MSB then XORs **adjacent binary bits**.
- Gray-to-Binary copies the MSB then uses a **cumulative XOR**.
- The binary **1011** converts to Gray **1110**.
- The Gray **1110** converts to binary **1011**.
- **ASCII** is a **7-bit** code representing **128** characters; extended ASCII uses **8 bits** for **256**.
- In ASCII, 'A' is **65**, 'a' is **97**, and '0' is **48**.
- The difference between an uppercase and lowercase ASCII letter is **32**.
- **EBCDIC** is an **8-bit** IBM code representing **256** characters, used on **mainframes**.
- A single **parity** bit detects **odd-numbered** bit errors but **cannot correct** any.
- Parity cannot detect an **even** number of bit errors.
- **Even parity** makes the total count of 1s even, and **odd parity** makes it odd.
- **Hamming code** can **detect 2-bit and correct 1-bit** errors.
- **Weighted** codes like **8421** and **2421** assign positional weights, while **non-weighted** codes like **Gray** and **Excess-3** do not.
- Self-complementing weighted codes include **2421** and **84-2-1**, whose weights sum to **9**.
- **8421 BCD** is weighted but **not** self-complementing, while **2421** is **both**.
- **Sequential** codes increase by one binary step each time; examples include **8421** and **Excess-3**.

Boolean Algebra

- Boolean algebra was formalized by **George Boole** and applied to switching circuits by **Claude Shannon** in **1938**.
- Boolean variables take exactly **2** values: **0** and **1**.
- The **3** fundamental Boolean operations are **AND (.)**, **OR (+)**, and **NOT (')**.
- Boolean operator precedence is **NOT > AND > OR**.
- The **identity** laws are **A+0=A** and **A.1=A**.
- The **null/dominance** laws are **A+1=1** and **A.0=0**.
- The **idempotent** laws are **A+A=A** and **A.A=A**.
- The **complement** laws are **A+A'=1** and **A.A'=0**.
- The **involution** law states **(A')' = A**.
- The **commutative** laws are **A+B=B+A** and **A.B=B.A**.
- The **associative** laws are **(A+B)+C=A+(B+C)** and **(A.B).C=A.(B.C)**.
- The **distributive** laws are **A.(B+C)=A.B+A.C** and **A+B.C=(A+B).(A+C)**.
- The second distributive form **A+BC=(A+B)(A+C)** is unique to **Boolean** algebra.
- The **absorption** laws are **A+A.B=A** and **A.(A+B)=A**.
- The absorption variants are **A+A'.B=A+B** and **A.(A'+B)=A.B**.
- The **consensus** theorem is **AB+A'C+BC = AB+A'C**, dropping the redundant **BC** term.
- The **2** De Morgan theorems are **(A.B)'=A'+B'** and **(A+B)'=A'.B'**.
- De Morgan's rule is to **break the bar and change the operator** (swap AND<->OR).
- The **duality principle** swaps **AND<->OR** and **0<->1** while keeping variables **unchanged**.
- **Duality** keeps variables as-is, but **De Morgan complement** also **negates** variables.
- A truth table for n variables has **2^n** rows.
- The number of distinct Boolean functions of n variables is **2^(2^n)**, e.g., **16** for 2 variables.
- A **minterm** is a product term using all variables and is true for exactly **1** input combination.
- A **maxterm** is a sum term using all variables and is false for exactly **1** input combination.
- **Canonical SOP** is the OR of **minterms** where the function equals **1**.
- **Canonical POS** is the AND of **maxterms** where the function equals **0**.
- In a **minterm** a variable is complemented if its bit is **0**; in a **maxterm** it is complemented if its bit is **1**.
- The relationship between minterm and maxterm is **m_i = (M_i)'**.
- To convert SOP to POS, take maxterms of the **missing** (complementary) indices.
- The example **F = Sigma m(0,2,5,7)** of 3 variables equals **Pi M(1,3,4,6)**.
- The complement F' in canonical form is **Sigma m** of the **remaining** minterm indices.

Logic Minimization

- **Algebraic simplification** is **not guaranteed** to reach the minimal form.
- A **K-map** arranges **2^n** cells in **Gray-code** order so adjacent cells differ by **1** variable.
- A 2-variable K-map has **4** cells, a 3-variable has **8**, and a 4-variable has **16**.
- K-map row/column labels follow Gray order **00, 01, 11, 10**.
- A K-map **wraps around**: top/bottom and left/right edges are adjacent.
- K-map groups must be **rectangular** and contain **1, 2, 4, 8, ...** (**powers of 2**) cells.
- A group of **2^k** cells eliminates **k** variables from the product term.
- A **prime implicant** is a group that **cannot be enlarged** further.
- An **essential prime implicant** covers at least **1** minterm covered by **no other** PI.
- A **don't-care** (X) is an input that **never occurs** or whose output **does not matter**.
- Don't-cares may be treated as **0 or 1**, whichever gives **larger** groups.
- For SOP minimization you group the **1s**; for POS minimization you group the **0s**.

- The rule of thumb is **largest groups, fewest groups**.
- All **EPIs** are **PIs**, but not all PIs are **essential**.
- **Quine-McCluskey** is a **tabular, algorithmic** method suited for **computer** implementation.
- Q-M combines terms differing in exactly **1** bit to find **prime implicants**.
- K-maps become impractical beyond about **6** variables, where **Q-M** is preferred.
- A **hazard** is a momentary **glitch** caused by unequal **propagation delays**.
- A **static-1 hazard** momentarily dips **1->0->1**, and a **static-0 hazard** momentarily rises **0->1->0**.
- A **dynamic hazard** makes **multiple** transitions when one was expected.
- Static hazards are cured by adding **redundant (consensus)** terms.
- A static-1 hazard relates to **SOP**, and a static-0 hazard relates to **POS**.

Logic Gates

- The **3** basic gates are **AND**, **OR**, and **NOT**.
- The AND gate outputs **1** only when **all** inputs are 1, expressed as **Y=A.B**.
- The OR gate outputs **1** when **any** input is 1, expressed as **Y=A+B**.
- The NOT gate outputs the **complement**, expressed as **Y=A'**.
- The **2** universal gates are **NAND** and **NOR**.
- **NAND** equals **(A.B)'** and **NOR** equals **(A+B)'**.
- A gate is **universal** because it can build **any** Boolean function alone.
- Building a NOT from NAND needs **1** gate (tie inputs together).
- Building an AND from NAND needs **2** gates.
- Building an OR from NAND needs **3** gates.
- Building an OR from NOR needs **2** gates, and AND from NOR needs **3** gates.
- The **XOR** gate outputs **1** when inputs **differ**, expressed as **A'B+AB'**.
- The **XNOR** gate outputs **1** when inputs are **equal**, expressed as **AB+A'B'**.
- XOR of n bits equals **1** when the number of 1s is **odd**.
- The XOR identities are **A XOR 0 = A**, **A XOR 1 = A'**, **A XOR A = 0**, and **A XOR A' = 1**.
- **XOR** is **not** a universal gate.
- A two-level **SOP** maps directly to a **NAND-NAND** circuit.
- A two-level **POS** maps directly to a **NOR-NOR** circuit.
- **Fan-in** is the number of inputs a gate accepts, and **fan-out** is how many inputs one output can drive.
- Standard TTL fan-out is about **10**.
- **TTL** uses **BJTs** while **CMOS** uses **MOSFETs**.
- **CMOS** has the lowest **static power** consumption.
- **ECL** is the **fastest** logic family but consumes the most **power**.

Combinational Circuits

- A **combinational** circuit output depends only on **present inputs**, with **no memory**.
- Combinational design has **5** steps: **problem statement, variable assignment, truth table, simplification, logic diagram**.
- A **half adder** has **2** inputs with **Sum=A XOR B** and **Carry=A.B**.
- A **full adder** has **3** inputs with **Sum=A XOR B XOR Cin** and **Cout=AB+BCin+ACin**.
- A full adder can be built from **2** half adders and **1** OR gate.
- The full-adder carry is the **majority** function of its 3 inputs.

- A **half subtractor** has **Difference=A XOR B** and **Borrow=A'.B**.
- A **full subtractor** has **Difference=A XOR B XOR Bin** and **Borrow=A'B+A'Bin+B.Bin**.
- The half-adder carry is **A.B** but the half-subtractor borrow is **A'.B**.
- A **ripple-carry adder** has worst-case carry delay **O(n)**.
- A **carry look-ahead adder** reduces carry delay to about **O(log n)**.
- In a CLA, **Generate G=A.B** and **Propagate P=A XOR B**, with carry **C(i+1)=G+P.C(i)**.
- A **multiplexer** is a **many-to-one** selector with **2^n** inputs, **n** selects, and **1** output.
- A **4:1 MUX** needs **2** select lines, and an **8:1 MUX** needs **3**.
- A **2^n:1 MUX** can implement any function of **n** variables directly.
- A **2^(n-1):1 MUX** implements any n-variable function by feeding **0,1,var,var'** on data lines.
- A **demultiplexer** is a **one-to-many** distributor: **1** input to **2^n** outputs via **n** selects.
- A **DEMUX** with its input tied to **1** behaves as a **decoder**.
- An **encoder** converts **2^n** inputs into an **n-bit** code.
- A plain encoder fails when **multiple** inputs are active simultaneously.
- A **priority encoder** resolves multiple active inputs by encoding the **highest priority** one.
- An **8-to-3** encoder has **8** inputs and **3** outputs.
- A **decoder** converts an **n-bit** code into **2^n** outputs, activating exactly **1**.
- Each decoder output equals one **minterm**.
- A decoder plus **OR** gate implements any **SOP** function.
- A **3-to-8** decoder has **3** inputs and **8** outputs.
- A **magnitude comparator** produces **3** outputs: **A>B**, **A=B**, and **A<B**.
- Equality (A=B) is detected using **XNOR** of each bit pair.
- A 1-bit comparator gives **A>B=A.B'** and **A<B=A'.B**.
- Binary-to-Gray conversion uses **G_i = B_i XOR B_(i+1)** with the MSB copied.
- Gray-to-Binary conversion uses a **cumulative XOR**.
- A **parity generator** uses a cascade of **XOR** gates.
- An **even-parity** bit is the **XOR** of all data bits, and **odd-parity** is its **complement (XNOR)**.
- A parity checker outputs **1** on error but only detects **odd-count** errors.

Latches and Flip-Flops

- A **latch** is **level-sensitive**, while a **flip-flop** is **edge-triggered**.
- A **NOR-based SR latch** has its forbidden state at **S=R=1**.
- A **NAND-based SR latch** is active-**low** with forbidden state at **S=R=0**.
- An SR latch holds its state when **S=R=0** (NOR type).
- A **gated SR latch** responds only when **enable=1**.
- A **D latch** removes the **invalid** state by making **R = D'**.
- A D latch is **transparent**: **Q follows D** while enabled.
- The **4** flip-flop types are **SR**, **D**, **JK**, and **T**.
- The **D** flip-flop has **Q(t+1)=D**.
- The **T** flip-flop has **Q(t+1)=T XOR Q** and **toggles** when T=1.
- The **JK** flip-flop has **Q(t+1)=JQ'+K'Q** and **toggles** when **J=K=1**.
- The **SR** flip-flop has **Q(t+1)=S+R'Q** and is invalid when **S=R=1**.
- The **JK** flip-flop resolves the SR **invalid** state by **toggling**.
- A **characteristic table** gives **Q(t+1)** from inputs and present state.
- An **excitation table** gives the **required inputs** for a desired transition.

- The JK excitation for transition 0->1 is **J=1, K=X**.
- The JK excitation for transition 1->0 is **J=X, K=1**.
- The T flip-flop needs **T=1** for any state change (0->1 or 1->0).
- The **JK** excitation table has the most **don't-cares**.
- Converting JK to D requires **J=D** and **K=D'**.
- Converting JK to T requires **J=K=T** (tie together).
- Converting D to T requires **D=T XOR Q**.
- A **level-triggered** device responds during the clock **level**, while **edge-triggered** responds at the clock **edge**.
- A **master-slave** flip-flop uses **2** latches on **opposite** clock phases.
- **Setup time** is the data-stable interval **before** the clock edge.
- **Hold time** is the data-stable interval **after** the clock edge.
- Violating setup/hold causes **metastability**.
- The **race-around** condition occurs in a **level-triggered JK** when **J=K=1**.
- Race-around is cured by **master-slave** or **edge-triggering**, or making the clock pulse **narrower than the propagation delay**.

Registers and Counters

- A **register** of n flip-flops stores an **n-bit** word.
- The **4** shift-register types are **SISO**, **SIPO**, **PISO**, and **PIPO**.
- **SIPO** performs **serial-to-parallel** conversion.
- **PISO** performs **parallel-to-serial** conversion.
- An n-bit SISO register needs **n** clock pulses to load serially.
- A **universal shift register** supports **4** operations: **shift-left, shift-right, parallel-load, hold**, chosen by **2** select lines.
- In an **asynchronous (ripple)** counter, each flip-flop is clocked by the **previous** FF's output.
- A ripple counter's worst-case delay is **n x t_pd**.
- An n-flip-flop ripple counter counts modulo **2^n**.
- In a **synchronous** counter, all flip-flops share the **same clock**.
- A synchronous counter's delay is just **1** flip-flop delay, independent of n.
- A **synchronous** counter needs **more** combinational logic than a ripple counter.
- A **mod-N** counter has **N** states and needs **ceil(log2 N)** flip-flops.
- A **decade counter** is a **mod-10** counter needing **4** flip-flops.
- A mod-10 counter resets at count **1010 (10)**.
- An **up/down** counter uses a **direction control** line.
- A **ring counter** of n flip-flops has **n** states and circulates a single **1**.
- A **Johnson counter** of n flip-flops has **2n** states using **complemented** feedback.
- A ring counter needs **no** output decoding, while a Johnson counter needs **minimal 2-input** decoding.
- Synchronous counter design has **5** steps: **state diagram, state table, FF excitation, K-map, circuit**.
- A counter's flip-flop count equals **ceil(log2(states))**.
- A counter must be **self-starting** so **unused states** return to the main sequence.

Finite State Machines

- The **2** types of finite state machines are **Mealy** and **Moore**.
- A **Mealy** machine output depends on **present state and input**.
- A **Moore** machine output depends on the **present state only**.

- In a Mealy machine outputs are labeled on **transitions**, while in Moore they are on **states**.
- A **Mealy** machine usually needs **fewer** states than an equivalent **Moore** machine.
- A **Moore** machine reacts to input **1 clock later** and is **glitch-free**.
- A **state diagram** has nodes as **states** and edges as **transitions**.
- A **state table** lists **present state, input, next state, output**.
- **State reduction** merges **equivalent** states having the same output and next-state for **all** inputs.
- The **3** common state-assignment encodings are **binary**, **Gray**, and **one-hot**.
- **One-hot** encoding uses **N** flip-flops for N states with the **simplest/fastest** logic.

Programmable & Memory Devices

- A **ROM** with n address lines has **2^n** words and acts as a **decoder + OR** array.
- A ROM has a **fixed AND** array and a **programmable OR** array.
- A **ROM** can realize any combinational function because it stores all **2^n** minterms.
- A **PLA** has **both** the AND and OR arrays **programmable**.
- A **PAL** has a **programmable AND** array and a **fixed OR** array.
- The **3** programmable devices ranked by flexibility are **PLA (most)**, **PAL**, and **ROM**.
- A **2^n:1 MUX** implements any n-variable function by placing **truth-table outputs** on data inputs.
- A **decoder + OR** implements any function by ORing the **minterm** lines where F=1.

Fixed and Floating Point

- **Fixed-point** representation keeps the **binary point** at a constant position.
- Fixed-point resolution (smallest step) is **2^(-f)** for f fraction bits.
- Fixed-point has a **limited dynamic range** compared to floating-point.
- **Floating-point** stores a number as **(-1)^S x mantissa x base^exponent**.
- A floating-point number has **3** fields: **sign**, **exponent**, and **mantissa**.
- **Normalization** puts the mantissa in the form **1.xxxx**.
- In IEEE-754 the leading **1** is **hidden/implicit**, giving 1 extra precision bit.
- The exponent is stored in **biased (excess)** form for easy comparison.
- **IEEE 754 single precision** is **32** bits: **1** sign, **8** exponent, **23** mantissa.
- **IEEE 754 double precision** is **64** bits: **1** sign, **11** exponent, **52** mantissa.
- The single-precision bias is **127**, and the double-precision bias is **1023**.
- The bias formula is **2^(exp_bits-1) - 1**.
- The normalized value formula is **(-1)^S x 1.M x 2^(E-bias)**.
- With the hidden bit, single precision has **24** effective mantissa bits and double has **53**.
- The actual exponent equals **stored exponent - bias**.
- **Overflow** in floating-point makes the result **infinity**.
- **Underflow** makes the result approach **zero** or become **denormal**.
- IEEE **infinity** has exponent all **1s** and mantissa **0**.
- IEEE **NaN** has exponent all **1s** and mantissa **nonzero**.
- IEEE **zero/denormal** has exponent all **0s**.
- Floating-point addition has **4** steps: **align exponents, add mantissas, normalize, round**.
- Floating-point multiplication **adds the exponents**, **subtracts one bias**, and **multiplies mantissas**.
- The default IEEE rounding mode is **round-to-nearest-even**.

Exam Focus Rapid Fire

- The **2's complement** of a number is **invert all bits + 1**.
- The overflow flag equals **Cin XOR Cout** at the **MSB**.
- A mod-N counter requires **ceil(log2 N)** flip-flops.
- A ring counter has **n** states and a Johnson counter has **2n** states.
- A **2^n:1** MUX has **n** select lines.
- The full-adder carry equals **AB+BCin+ACin**.
- The IEEE single-precision bias is **127** and double is **1023**.
- Gray code changes exactly **1** bit per step.
- A K-map group of **2^k** cells removes **k** variables.
- Unsigned n-bit range is **0 to 2^n - 1**; signed 2's-complement is **-2^(n-1) to +2^(n-1)-1**.
- For SOP grouping use **1s**, for POS grouping use **0s**.
- **NAND** and **NOR** are the **2** universal gates; **XOR** is not.
- **Mealy** output depends on **state+input**; **Moore** output depends on **state only**.
- A **half adder** has **no carry-in**; a **full adder** has **3** inputs.
- **PLA** has both arrays programmable; **PAL** has only the **AND** array programmable.

Supplementary One-Liners: Number Systems

- The decimal value of binary **1111** is **15**, the maximum for **4** bits.
- The decimal value of binary **10000** is **16**, needing **5** bits.
- The octal **17** equals binary **001 111** = **1111** = decimal **15**.
- The hexadecimal **FF** equals decimal **255** and binary **11111111**.
- The hexadecimal **10** equals decimal **16**.
- The hexadecimal **2AF** equals decimal **687**.
- The binary **1100** equals octal **14** and hexadecimal **C**.
- A **32-bit** word holds **8** hexadecimal digits.
- A **16-bit** word holds **4** hexadecimal digits or **2** bytes.
- The number of bits to represent decimal N is **ceil(log2(N+1))**.
- The 9's complement of a decimal digit d is **9 - d**.
- The 10's complement equals **9's complement + 1**.
- The r's complement of an n-digit number is **r^n - number**.
- The (r-1)'s complement of an n-digit number is **r^n - number - 1**.
- The 1's complement is the **(r-1)'s complement** for base **2**.
- The 2's complement is the **r's complement** for base **2**.
- The 7's complement is used in **octal** and the 15's complement in **hexadecimal**.
- The decimal **-5** in 8-bit 2's complement is **11111011**.
- The decimal **-1** in 8-bit 2's complement is **11111111** (all ones).
- The decimal **-128** in 8-bit 2's complement is **10000000**.
- A sign-extended 4-bit **1011** to 8 bits becomes **11111011**.
- The MSB weight in an n-bit 2's-complement number is **-2^(n-1)**.

Supplementary One-Liners: Codes and Boolean

- The BCD for decimal **0** is **0000** and for **9** is **1001**.

- The Excess-3 for decimal **0** is **0011** and for **9** is **1100**.
- The Gray code for decimal **7** is **0100**.
- The number of invalid BCD codes is **6** (1010 through 1111).
- A **checksum** and **CRC** are error-detecting techniques beyond simple **parity**.
- **Two-dimensional parity** can **correct** a **single-bit** error.
- The number of De Morgan theorems is **2**.
- The Boolean expression **A+A'B** simplifies to **A+B**.
- The Boolean expression **A(A+B)** simplifies to **A**.
- The Boolean expression **AB+AB'** simplifies to **A**.
- The Boolean expression **(A+B)(A+B')** simplifies to **A**.
- The expression **A XOR A** equals **0** and **A XNOR A** equals **1**.
- The dual of **A+1=1** is **A.0=0**.
- The complement of **AB+CD** is **(A'+B')(C'+D')**.
- The number of minterms in an n-variable function is up to **2^n**.
- The sum of all minterms of n variables equals **1**.
- The product of all maxterms of n variables equals **0**.
- For 3 variables, minterm **m5** corresponds to **AB'C** (binary 101).
- For 3 variables, maxterm **M5** corresponds to **A'+B+C'** (binary 101).

Supplementary One-Liners: K-maps and Gates

- A 4-variable K-map can have a maximum group of **16** cells, which equals **F=1** (constant).
- A group of **8** cells in a 4-variable K-map leaves a **single-literal** term.
- A group of **1** cell in a 4-variable K-map gives a **4-literal** product term.
- The four corners of a K-map can form a single group of **4** due to **wrap-around**.
- The minimum number of gates to implement XOR using NAND is **4**.
- A 2-input XOR requires **4** NAND gates or **5** basic gates.
- The number of select lines for a **16:1** MUX is **4**.
- The number of outputs of a **4-to-16** decoder is **16**.
- To build a **32:1** MUX from **8:1** MUXes you need **4** plus a **4:1** MUX.
- A **1:4** DEMUX has **2** select lines.
- The number of half adders needed to build a full adder is **2**.
- An n-bit ripple-carry adder uses **n** full adders.
- A **4-bit** parallel adder needs **4** full adders.
- A BCD adder uses a **4-bit** binary adder plus correction logic adding **6**.
- A **4-bit** magnitude comparator compares two **4-bit** numbers.

Supplementary One-Liners: Sequential Depth

- The number of flip-flops in a **mod-16** counter is **4**.
- The number of flip-flops in a **mod-6** counter is **3**.
- A **mod-12** counter needs **4** flip-flops.
- A **4-bit** ring counter has **4** states.
- A **4-bit** Johnson counter has **8** states.
- A **3-bit** ripple counter counts from **0 to 7**.
- The maximum count of an n-bit binary counter is **2^n - 1**.

- The number of states in a **decade** counter is **10**.
- A T flip-flop with T held at **1** acts as a **frequency divider by 2**.
- A chain of n toggle flip-flops divides frequency by **2^n**.
- The SR flip-flop characteristic equation is **Q+ = S + R'Q** with constraint **SR=0**.
- The number of don't-cares in the JK 0->0 transition is **1** (K=X).
- A D flip-flop is also called a **delay** flip-flop.
- The number of latches in a master-slave flip-flop is **2**.
- A **positive edge** triggers on the **0->1** clock transition.
- A **negative edge** triggers on the **1->0** clock transition.
- Converting a D flip-flop to JK uses the equation **D = JQ' + K'Q**.
- The propagation delay of a synchronous counter is independent of the number of **flip-flops**.

Supplementary One-Liners: Arithmetic and Mixed

- The IEEE single-precision exponent range (unbiased) is **-126 to +127**.
- The IEEE double-precision exponent range (unbiased) is **-1022 to +1023**.
- The stored exponent **255** (all 1s) in single precision signals **infinity or NaN**.
- The smallest positive normalized single-precision value uses exponent **1** (biased).
- The decimal **0.0** in IEEE-754 has all **0** bits (positive zero).
- The number **-0.0** differs from **+0.0** only in the **sign** bit.
- The mantissa of **1.0** in IEEE-754 is all **0s** (with hidden 1).
- A **32-bit** fixed-point can be split as **16 integer + 16 fraction** bits (Q16.16).
- The dynamic range of floating-point is **larger** than fixed-point for the same bit width.
- The number of distinct values in any n-bit format is always **2^n**.
- The decimal **10** in binary is **1010**.
- The decimal **100** in binary is **1100100**.
- The decimal **255** needs **8** bits and **256** needs **9** bits.
- The result of **1010 AND 0110** is **0010**.
- The result of **1010 OR 0110** is **1110**.
- The result of **1010 XOR 0110** is **1100**.
- The NOT of **1010** (4-bit) is **0101**.
- The number of rows in a 4-variable truth table is **16**.
- The number of cells in a 5-variable K-map is **32**.
- A 5-variable K-map is drawn as **2** overlaid 4-variable maps.
- The total Boolean functions of **3** variables number **256** (2^8).
- The minimum SOP and minimum POS of a function may differ in **gate count**.
- The consensus term of **AB + A'C** is **BC**.
- The number of literals in **A'BC + AB'** is **5**.
- A two-level logic circuit has a propagation delay of **2** gate delays.
- The fan-out limit prevents excessive **loading** of a gate output.
- The complement of a **minterm** is the corresponding **maxterm**.
- A function and its complement together cover all **2^n** minterms.
- The number of product terms in canonical SOP equals the number of **1s** in the truth table.
- The number of sum terms in canonical POS equals the number of **0s** in the truth table.

Supplementary One-Liners: Additional Recall

- The base of the binary system is **2**, chosen for **two-state** electronic devices.
- A **kilobyte** is **2^10 = 1024** bytes in binary measure.
- A **megabyte** is **2^20** bytes and a **gigabyte** is **2^30** bytes.
- The radix point separates the **integer** and **fractional** parts.
- The weight of the first fractional binary position is **2^-1 = 0.5**.
- A signed magnitude **8-bit** number ranges from **-127 to +127**.
- The 1's complement of **0110** is **1001**.
- The 2's complement of **0110** is **1010**.
- The addition **0101 + 0011** equals **1000** (5+3=8).
- A **NAND** gate with both inputs tied acts as a **NOT** gate.
- A **NOR** gate with both inputs tied acts as a **NOT** gate.
- The AND gate is the **multiplication** operation in Boolean algebra.
- The OR gate is the **addition** operation in Boolean algebra.
- A **buffer** gate outputs the **same** logic value as its input.
- A **tri-state** buffer has outputs **0, 1, and high-impedance Z**.
- The **enable** input of a tri-state buffer controls the **Z** state.
- A **2-input** AND gate has **4** truth-table rows.
- The number of NAND gates to realize **Y=AB+CD** is **3**.
- A half adder cannot handle a **carry-in**, unlike a full adder.
- The carry-out of a full adder is fed to the next stage's **carry-in**.
- An **n:1** MUX implements a function of **log2(n)** select variables.
- A decoder with **enable** can act as a **demultiplexer**.
- The number of AND gates inside an **n-to-2^n** decoder is **2^n**.
- A **priority encoder** adds a **valid** output to flag the **no-input** case.
- The **set** input of an SR latch forces output to **1**.
- The **reset** input of an SR latch forces output to **0**.
- A flip-flop stores exactly **1** bit of information.
- An n-bit register stores **n** bits and **2^n** possible values.
- A shift-right operation by 1 divides an unsigned number by **2**.
- A shift-left operation by 1 multiplies an unsigned number by **2**.
- An arithmetic right shift preserves the **sign** bit.
- The **state** of a sequential circuit is held in its **flip-flops**.
- The number of states reachable by m flip-flops is **2^m**.
- A Moore machine output is **stable** for the entire clock period.
- A Mealy machine may produce a **transient** output glitch.
- The **clock** synchronizes state changes in a synchronous circuit.
- The **frequency** of a clock is the inverse of its **period**.
- A **50% duty cycle** clock is high for half its period.
- The number of programmable arrays in a **PLA** is **2** (AND and OR).
- A **ROM**'s content is set during **manufacturing or programming** and is **non-volatile**.
- The address decoder of a **1K** ROM has **10** address lines.
- The number of words in a ROM with 12 address lines is **4096**.
- IEEE-754 reserves exponent **all-zeros** for **zero and denormals**.
- A denormalized number has an **implicit leading 0** instead of 1.
- Rounding **half-to-even** reduces statistical **bias** in repeated operations.
- The number of guard/round/sticky bits used in FP rounding is **3**.

Supplementary One-Liners: Extra Coverage

- The binary **110110** equals decimal **54**.
- The binary **0.11** equals decimal **0.75**.
- The octal **0.4** equals binary **0.100** and decimal **0.5**.
- The hexadecimal **0.8** equals decimal **0.5**.
- The decimal **45** equals octal **55** and hexadecimal **2D**.
- The decimal **200** equals binary **11001000** and hexadecimal **C8**.
- The largest number in **3** octal digits is **777** = decimal **511**.
- The largest number in **2** hex digits is **FF** = decimal **255**.
- Adding **1** to the all-ones n-bit number gives **0** with a **carry-out** (overflow in unsigned).
- The **weight** of bit position i in binary is **2^i**.
- The **2421** code for decimal **5** is **1011**.
- The Gray code sequence for 2 bits is **00, 01, 11, 10**.
- A **reflected** code is another name for **Gray** code.
- The number of bits that change from Gray **011** to **010** is **1**.
- The **ASCII** code for space is **32**.
- The **ASCII** code for 'Z' is **90** and 'z' is **122**.
- The Boolean function **F=A** is independent of **B**.
- The **XOR** of an even number of 1s is **0**.
- The **XNOR** is also called the **equivalence** or **coincidence** gate.
- The number of essential prime implicants can be **0** when all PIs overlap fully.
- A function with no essential PIs may have **multiple** minimal solutions.
- The **cyclic** property of Gray code means the last and first codes also differ by **1** bit.
- The number of literals saved by combining two minterms is **1** variable.
- A **tautology** is a Boolean function that is always **1**.
- A **contradiction** is a Boolean function that is always **0**.
- The number of inputs to a **majority** gate that must be 1 for output 1 is **more than half**.
- The full-adder sum is **1** when an **odd** number of its 3 inputs are 1.
- A **carry-save** adder is used in fast **multiplication**.
- The number of product terms a **PAL** can OR per output is **fixed**.
- A **FPGA** uses **look-up tables (LUTs)** to implement logic.
- The number of memory cells in a **4x4** RAM is **16**.
- A **static RAM (SRAM)** cell uses **6** transistors and needs **no refresh**.
- A **dynamic RAM (DRAM)** cell uses **1** transistor and **1** capacitor and needs **refresh**.
- The number of address lines for **1024** memory locations is **10**.
- The number of data lines for a byte-wide memory is **8**.
- A **multiplexer** is also known as a **data selector**.
- A **demultiplexer** is also known as a **data distributor**.
- The output of a decoder is **active-high** or **active-low** depending on design.
- A **seven-segment** display decoder converts BCD to **7** segment signals.
- A **BCD-to-seven-segment** decoder drives segments labeled **a through g**.
- The number of flip-flops to store one decimal digit in BCD is **4**.
- A **Schmitt trigger** adds **hysteresis** to clean noisy signals.
- The number of stable states in a **bistable** multivibrator is **2**.
- A **monostable** multivibrator has **1** stable state.

- An **astable** multivibrator has **0** stable states and acts as an **oscillator**.
- The clock for a sequential circuit is often generated by an **astable** multivibrator.
- The number of flip-flops changing simultaneously in a synchronous counter can be **multiple**.
- A **glitch** in a ripple counter output arises from **unequal** flip-flop delays.
- The maximum frequency of a counter is limited by its **propagation delay**.
- The setup-plus-hold window is the **aperture** time of a flip-flop.
- The product (-1)^S applies the **sign** in IEEE-754.
- The exponent field width determines the **range**, and the mantissa width determines the **precision**.
- Single precision has about **7** decimal significant digits.
- Double precision has about **15-16** decimal significant digits.
- The hidden bit makes the mantissa value lie in **[1, 2)** for normalized numbers.
- An IEEE-754 number with biased exponent **0** and nonzero mantissa is a **subnormal**.
- The **round** step in FP addition may cause a second **normalization**.
- The smallest representable gap between consecutive floats is the **ULP (unit in the last place)**.
- The number of bytes in a single-precision float is **4**, and in double-precision is **8**.
- A **2's complement** adder/subtractor selects subtraction by feeding a **1** to the carry-in and **inverting** B.
- The carry-in to the LSB during 2's-complement subtraction is **1**.
- The number of full adders plus XOR gates in an n-bit adder/subtractor is **n** each.
- A **barrel shifter** can shift by multiple positions in **1** step.
- The number of select lines in a barrel shifter for n positions is **log2 n**.
- The result of sign-extending **0xF** (4-bit -1) to 8 bits is **0xFF**.
- The number of guard bits minimum for correct FP rounding is **1**, plus round and sticky for accuracy.